
Working title. *Muscle Memory for Robot Policies: Caching a VLA’s Own Experience in Its Own Representations.*

Outline (v0.2). Page budgets follow the 9-page main-text limit. The current draft policy is to write every section in full and cut at writing time; the budget is exceeded deliberately. Experiment codes (E0–E12) refer to the experiment list.

1. **Introduction** (~1.25 pp; Fig. 1). (i) Deployed robots repeat themselves while VLAs recompute every step; serving systems schedule this demand, none of them remove it. (ii) The central question: how much of closed-loop VLA inference is redundant, in the inter-frame-compression sense? (iii) Key distinction from prior retrieval caches: keys are extracted from the teacher’s own forward pass rather than from an external encoder. (iv) System summary: training-free threshold dispatch over a library of the teacher’s own successes. (v) Three contributions: (a) definition and first closed-loop measurement of recoverable redundancy $R(\varepsilon)$, including its temporal structure; (b) the serving primitive, orthogonal to distillation, quantization, and step-reduction; (c) the frontier and its economics (E1 numbers; cross-scene transfer **TODO**). Fig. 1 combines the system diagram, the four-stage lifecycle, and an episode hit-strip.
2. **Related work** (~0.6 pp). Six families: robot retrieval caches (RT-Cache; key provenance is the distinction); VLA acceleration primitives (orthogonal); the 2026 adaptive-computation line (SkiP, ElegantVLA: mechanisms without a defined quantity or a library); policy serving (ROSA, Armory: scheduling versus removal of demand); the cache lineage across domains (exact KV/prefix reuse, approximate semantic reuse, closed-loop approximate reuse here); concept ancestors (event-triggered control; uncertainty quantification and failure detection). Closes with the first-claim: to our knowledge, the first formal definition and closed-loop measurement of the recoverable redundancy of VLA inference; the systems-level distinction is stated structurally (keys before the backbone; concurrent caches key on backbone outputs and must run it every step).
3. **Recoverable redundancy** (~0.75–1 pp). The formalism of the companion note: step-level definition, failure of composition, $R_C(\varepsilon)$, the information-set stratification, attribution, monotonicity, the intrinsic limit. Full proofs and extended remarks in Appendix A. Three interfaces to the rest of the paper are stated explicitly: calibration solves Eq. (3) empirically; the E1 sweep is the empirical Pareto frontier of the threshold class; the attribution quadruple organizes the experimental chapters.
4. **System** (~1.5 pp). Key extraction (tap point, layout, pooling; the corresponding tap point in a second architecture); library construction under a success filter; retrieval with per-field normalization and weighted fusion; threshold calibration as the empirical program of Eq. (3); the three execution tiers (replay; warm start as experience-guided step reduction; full inference); the deployment lifecycle (collect, shadow, calibrate, run and append); the cost account (E0: hit $\approx$ 14 ms against 72 ms, the IR floor).
5. **Measuring $R(\varepsilon)$** (~1.5 pp; Fig. 2, Fig. 3, Table 1). (5.1) Main frontier: four panels, two benchmarks $\times$ two teachers, each panel with the threshold sweep (E1) and the oracle arm (E2 **TODO**) overlaid; teacher anchors marked. (5.2) Library-size scaling (E3). (5.3) Temporal structure (E4), one half-page: stickiness, run lengths, lockstep; one sentence on score discriminability (E5 **TODO**). (5.4) Key-source ablation (E7a **TODO**): internal representations against external encoders, the direct evidence for the title claim; extraction depth and fusion (E7b, E7c) remain in Appendix E. Table 1: pre-registered operating points with success rate, replay share, GPU-time per step, robots-per-GPU (1/IR), and resident memory; anchored by the serving mini-bench (E11 **TODO**).
6. **Does experience transfer?** (~0.75–1 pp; Fig. 4). Cross-scene transfer on RoboCasa365 (E6 **TODO**): library built in one kitchen, queried in held-out kitchens, two teachers; analysis stratified by task type (preliminary: large-motion tasks transfer, precise dynamic tasks do not). Fall-back if the data are not ready: the main figure retreats to LIBERO and this section absorbs all RoboCasa material.
7. **Real-robot validation** (~0.3–0.5 pp; E13 **TODO**). The full deployment lifecycle executed on a physical SO-101 arm: collect, shadow, calibrate, serve. Three tasks stratified by motion scale

(drawer or lid, pick-and-place, insertion), approximately 330 episodes, paired evaluation at two operating points. Reports success rate, replay share, and end-to-end latency, where a hit is a local replay and a miss is a network round trip plus inference; the latency column instantiates the per-robot axis with physical measurements. Positioned as an existence proof, not a main result.

8. **Alternatives** (~ 0.75 pp; Table 2). (7.1) Distilled students: the regime map; the acquisition-cost column; the success-rate loss reported without qualification (E9). (7.2) Trained routers: counterfactual labels, retraining coupling to the library, the threshold as a byproduct of retrieval; the comparison E10. (7.3) Cheaper teachers: orthogonality and composability; warm start against blind step reduction (E8); appendix for the three-way control.
9. **Limitations** (~ 0.3 pp). Real-robot validation is small-scale and serves as an existence proof; quasi-static tasks (staleness sensitivity of replay); single embodiment class; the robots-per-GPU projection is an idealization anchored at one measured point; library growth is append-only with no eviction policy.
10. **Conclusion** (~ 0.1 pp).

Floats. Fig. 1 system, lifecycle, hit-strip. Fig. 2 main frontier, four panels with oracle overlay. Fig. 3 library-size scaling and temporal structure. Fig. 4 cross-scene transfer. Fig. 5 real-robot latency and success (E13). Table 1 operating points and economics. Table 2 alternatives. One comparison table in Section 2.

Appendix plan. A: full formalism, proofs, extended remarks. B: cost accounting, serving mini-bench (E11), warm-start three-way control (E8). C: history-augmented negatives, one table (E12). D: calibration protocol, pre-registration records, statistics. E: per-task breakdowns, key-space ablations E7b (depth) and E7c (fusion), score discriminability (E5). F: implementation, reproducibility, data and code release. A further appendix section, *Design choices and alternatives considered*, answers anticipated questions in declarative form (learned routers, distillation, memorization of the evaluation set, unbounded library growth, safety of replayed actions), restricted to questions with substantiated answers.

Remaining collection. E2 (oracle arm), E11 (serving mini-bench), E8 refresh, E13 (SO-101; teacher fine-tuning is the prerequisite); E1 RoboCasa cells, E6, and E10 in progress; E5 and E7 offline.